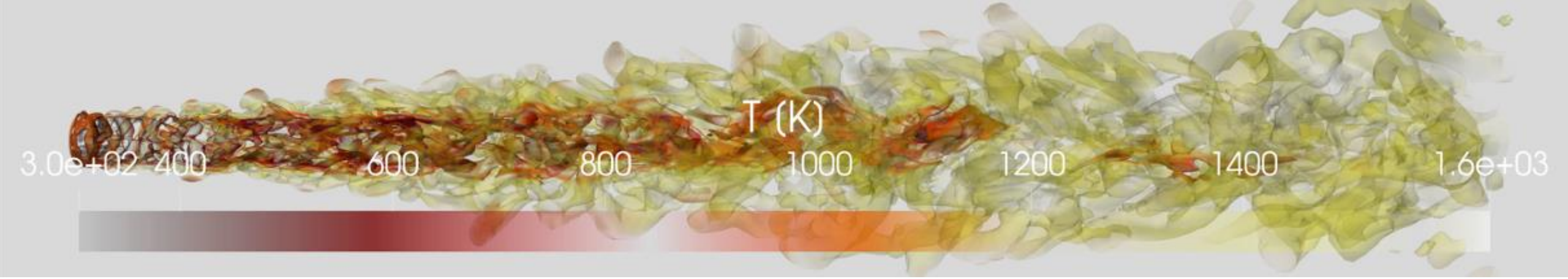

CYPHER data challenge: Benchmarking ML-enhanced turbulent combustion closures

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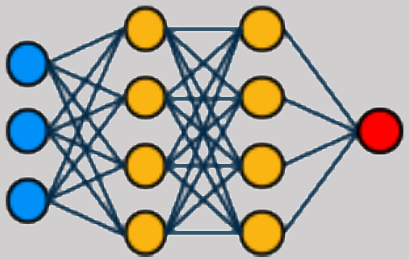
Funded by the European Union

CYPHER Cost Action objective

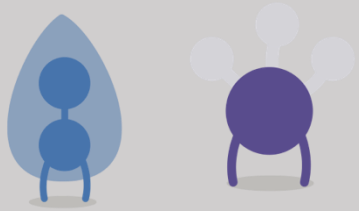
" Propel the **collaborations** between European researchers and industrial stakeholders to foster the use of **digital-physical** systems and ultimately promote a safe and sustainable adoption of **renewable synthetic fuels** to decarbonise **energy-intensive industries**"

1. Collaboration

2. Machine Learning



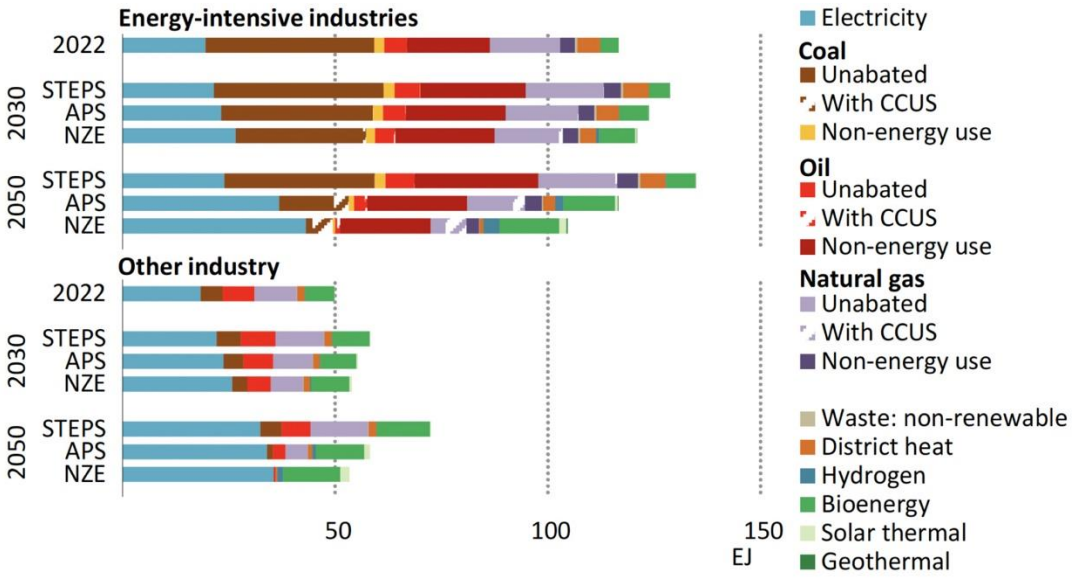
3. Renewable fuels

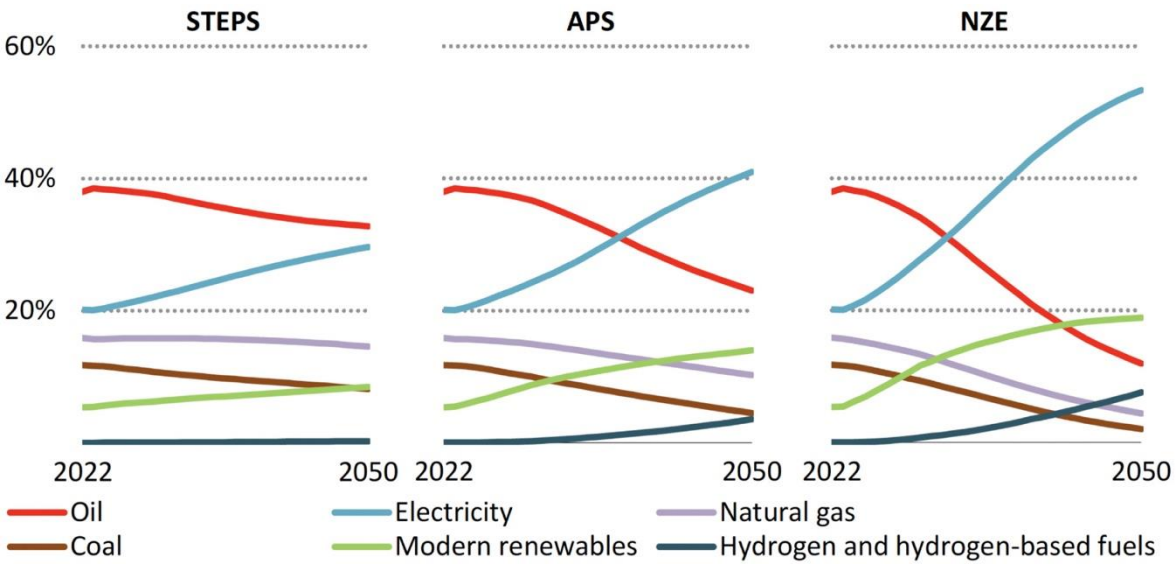


4. Energy-intensive industries

Combustion will play an important role in the energy transition

At least 40% of the world's energy supply will be based on hydrogen, hydrogen-based fuels, and natural gas [1].





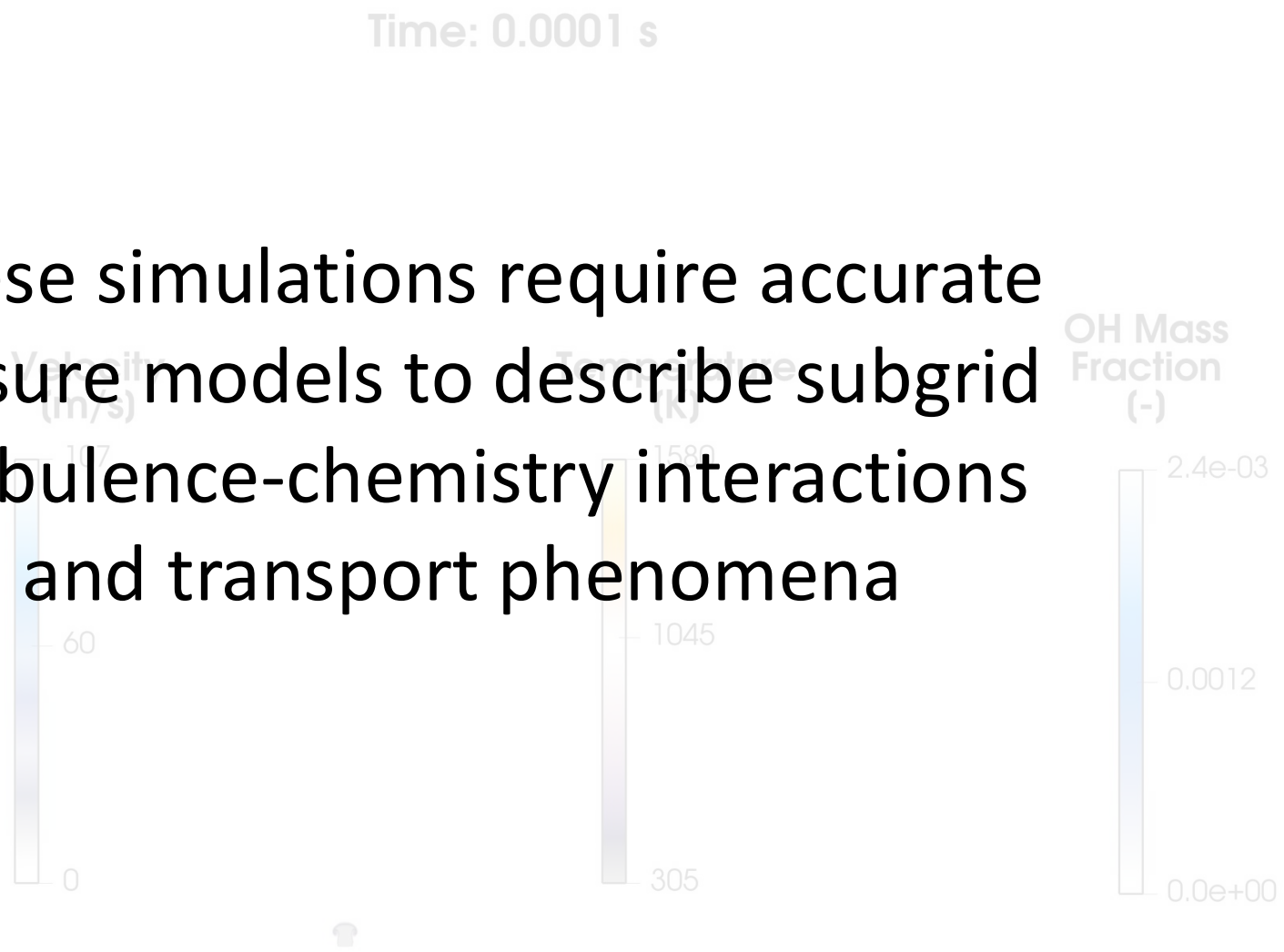
The vast majority of the energy request comes from the *hard-to-abate* or *energy-intensive* industries [2].

These industries need **high-temperature heat**, which is difficult to electrify and hence has to be provided in large parts by **combustion** processes

[1] Pitsch, H. (2024). *The transition to sustainable combustion: Hydrogen- and carbon-based future fuels and methods for dealing with their challenges*. *Proceedings of the Combustion Institute*, **40**(1–4), Article 105638. https://doi.org/10.1016/j.proci.2024.105638

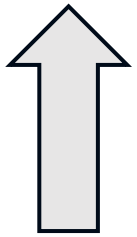
[2] U.S. Energy Information Administration, International energy outlook 2023, 2023, https://www.eia.gov/outlooks/ieo/.

High-fidelity simulations are becoming a fundamental tool to gain insights into complex combustion systems

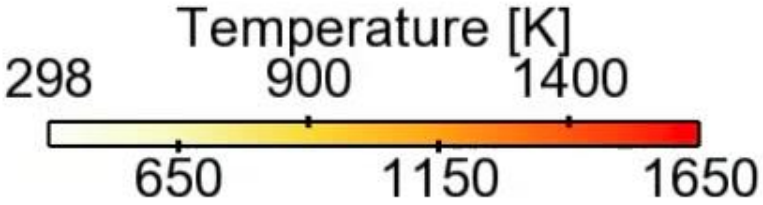


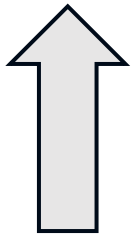
These simulations require accurate closure models to describe subgrid turbulence-chemistry interactions and transport phenomena

Hydrogen is a seemingly simple fuel, but modeling differential diffusion can be challenging even for laminar flames



Fresh unburnt gases

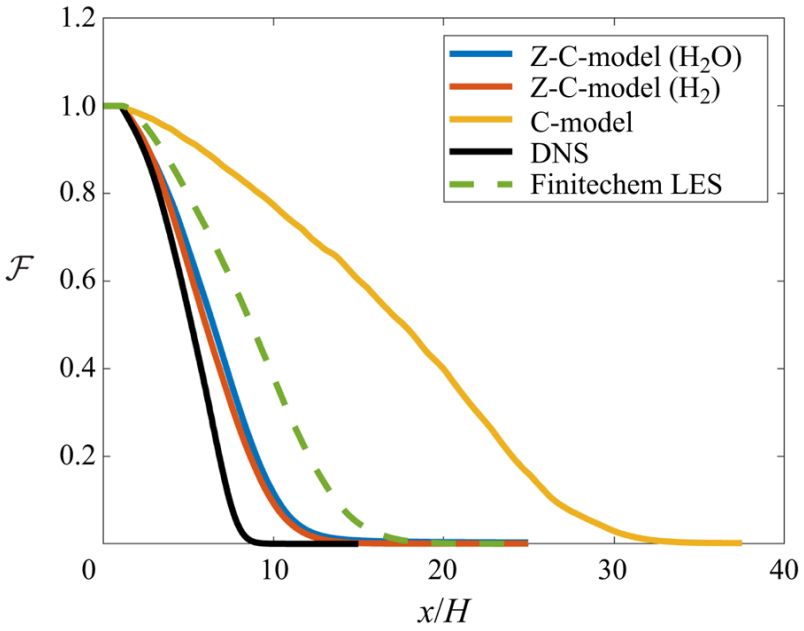


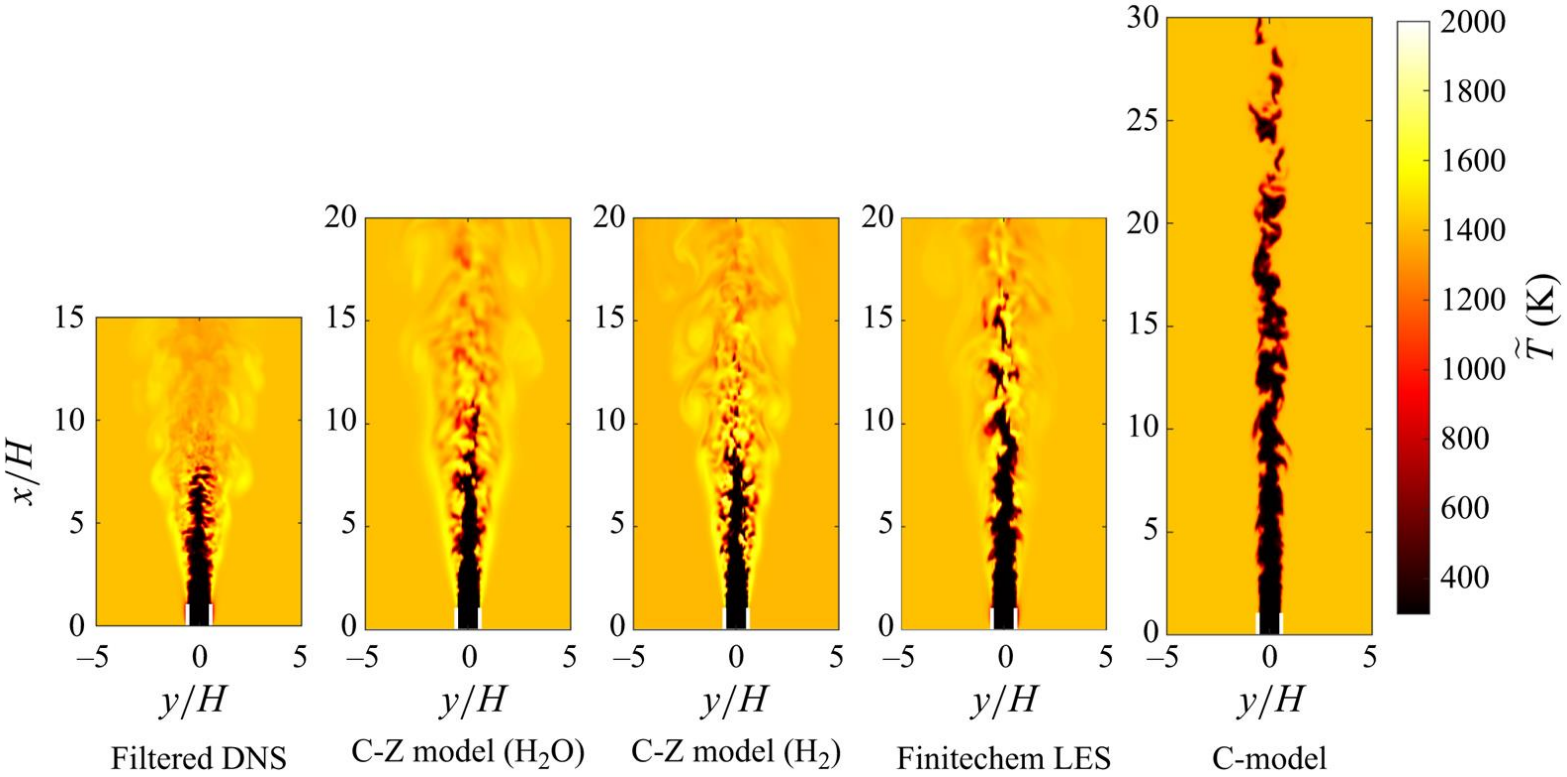


Fresh unburnt gases

[3] Berger, L., Kleinheinz, K., Attili, A., & Pitsch, H. (2019). *Characteristic patterns of thermodiffusively unstable premixed lean hydrogen flames*. *Proceedings of the Combustion Institute, 37*(2), 1879–1886. https://doi.org/10.1016/j.proci.2018.06.072

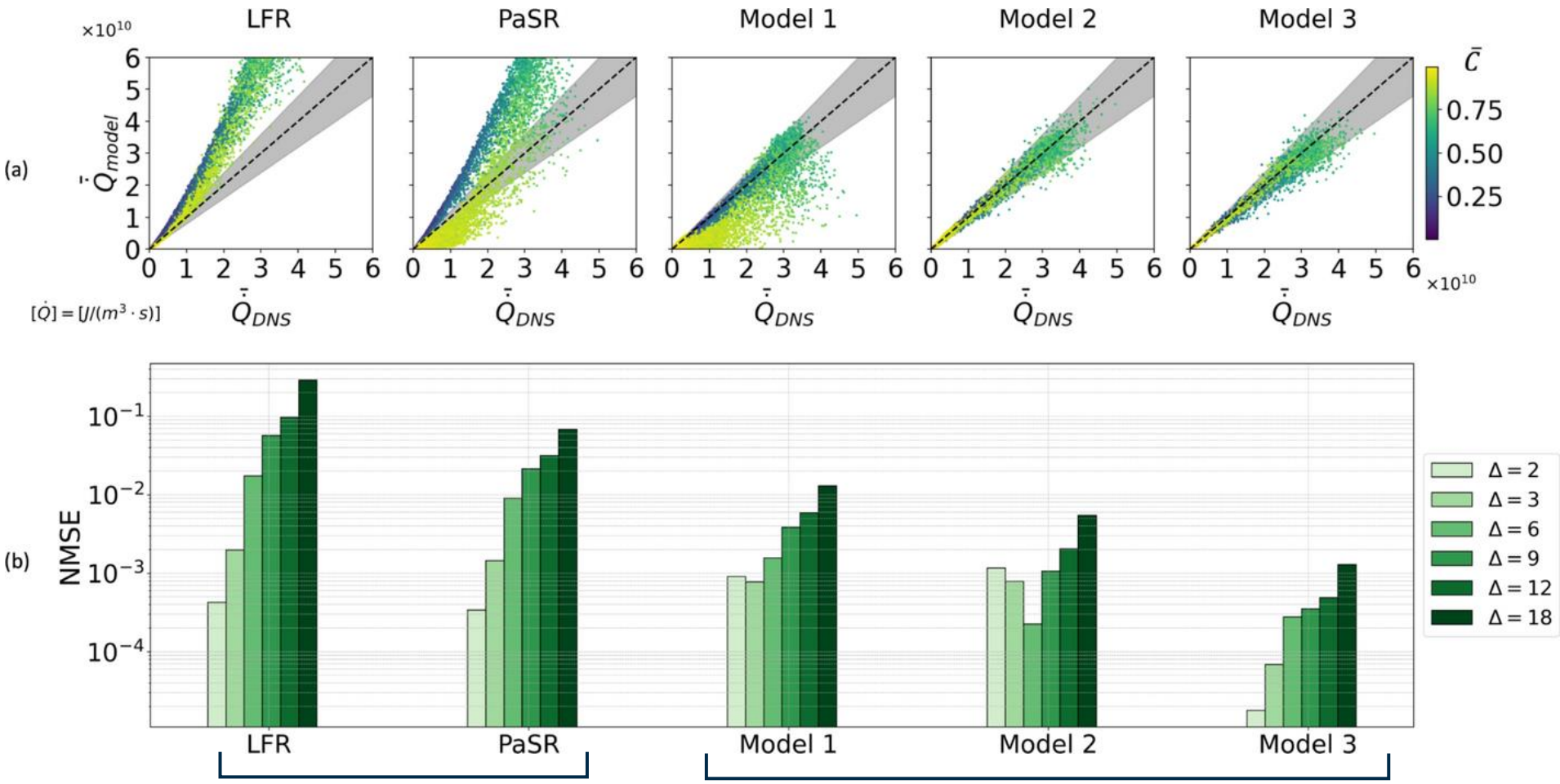
Neglecting thermo-diffusive instabilities can lead to macroscopic errors in the simulations





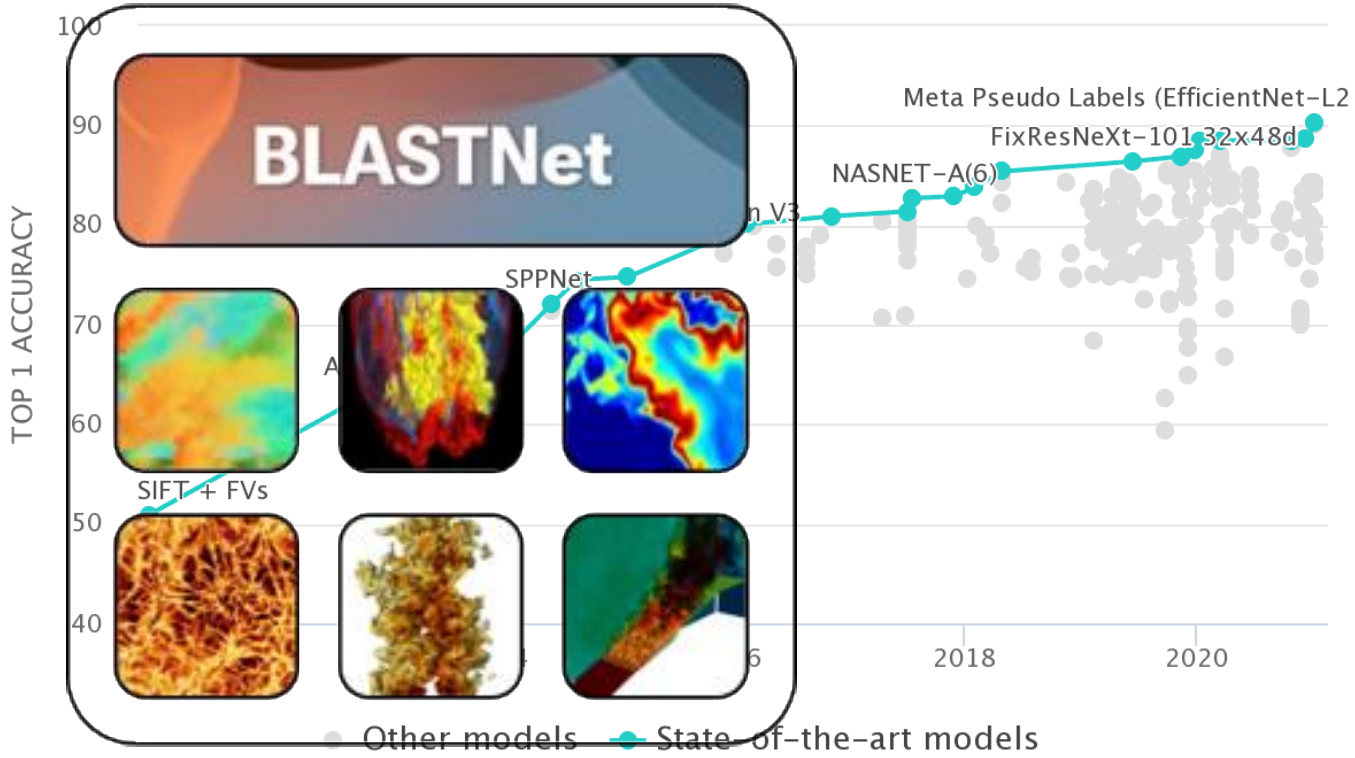
[4] Berger, L., Attili, A., Gauding, M., & Pitsch, H. (2025). *LES combustion model for premixed turbulent hydrogen flames with thermodiffusive instabilities: a priori and a posteriori analysis*. Journal of Fluid Mechanics, 1003, A33. doi:10.1017/jfm.2024.1201

Machine learning was shown potential to enhance existing turbulent combustion closures in LES



[5] Piu, L., Péquin, A., Freitas, R. S. M., Iavarone, S., Pitsch, H., & Parente, A. (2025). *A data-driven approach to refine the partially stirred reactor closure for turbulent premixed flames*. Flow, Turbulence and Combustion, 115(3), 1235–1260. https://doi.org/10.1007/s10494-024-00626-3

Shared databases enabled fast and standardized progress in machine learning, especially image recognition



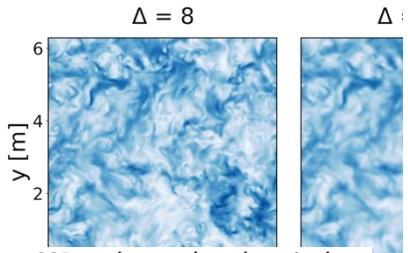
- Data to be **processed** (e.g. sub-filter closures)
- Large dimensions (up-to **100s GB** per simulation timestep)

- **Ready to use**, labeled data
- Easy to handle, **accessible** dataset

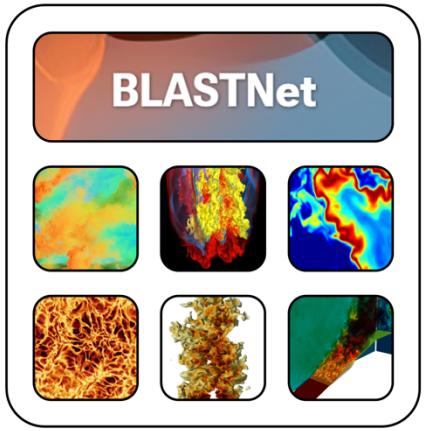
[6] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). *ImageNet: A large-scale hierarchical image database*. In Proceedings of the 2009 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
[7] Chung, W. T., Jung, K. S., Chen, J. H., & Ihme, M. (2022). *BLASTNet: A call for community-involved big data in combustion machine learning*. Appl. in En. and Comb. Sc.

Processing Direct Numerical Simulations (DNS) can be challenging from a practical standpoint

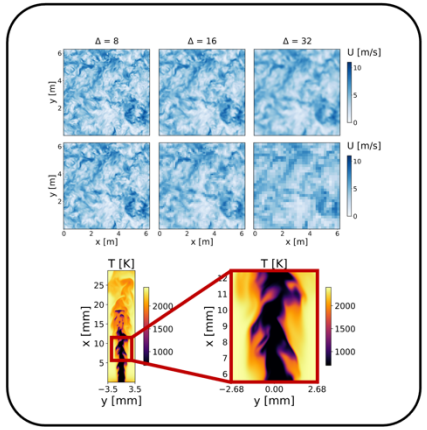
Documentation website



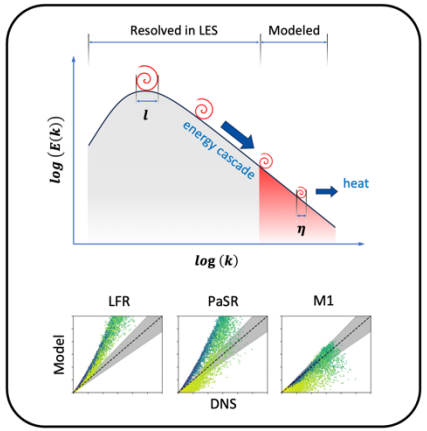
1. Handling large datasets from open-source repositories



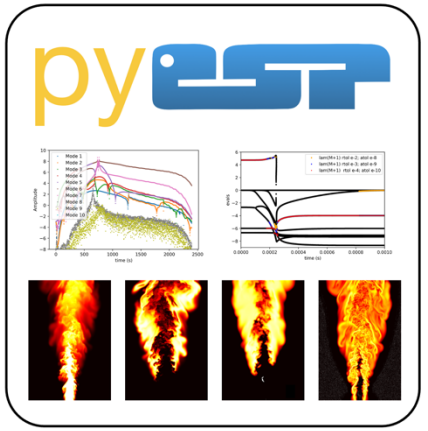
2. Filtering, downsampling, and sub-domain extraction

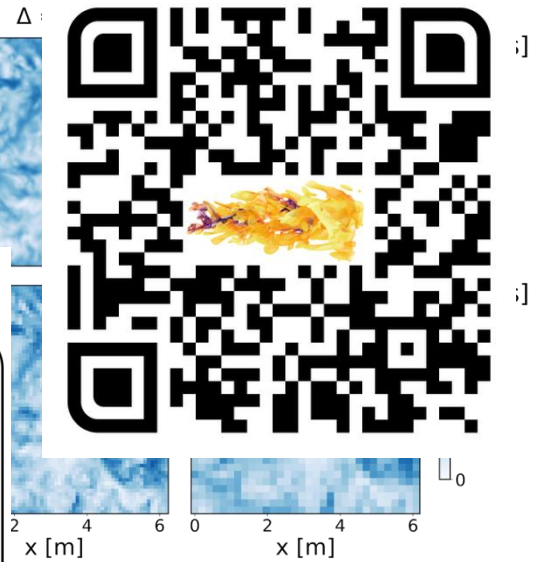


3. Sub-filter quantities and model assessment



4. CSP and complex chemical analyses for reactive flows





- Large dimensions (up-to **100s GB** per simulation timestep)

ıb-filter data to model
ıachine learning

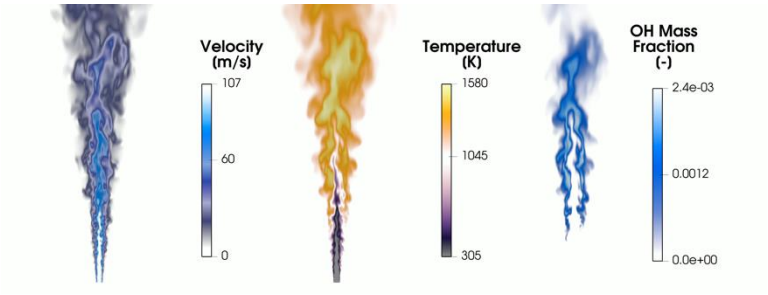
- Easy to handle, **accessible** datasets

[6] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). *ImageNet: A large-scale hierarchical image database*. In Proceedings of the 2009 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
[7] Chung, W. T., Jung, K. S., Chen, J. H., & Ihme, M. (2022). *BLASTNet: A call for community-involved big data in combustion machine learning*. Appl. in En. and Comb. Sc.

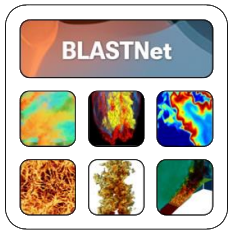
Brief recap

1. **CYPHER** cost action tackles digitalization to develop renewable synthetic fuels usage in the **hard-to-abate industries**;

2. **LES** are a powerful predictive tool, and machine learning can improve sub-filter **closure models** for **hydrogen-based** fuels;



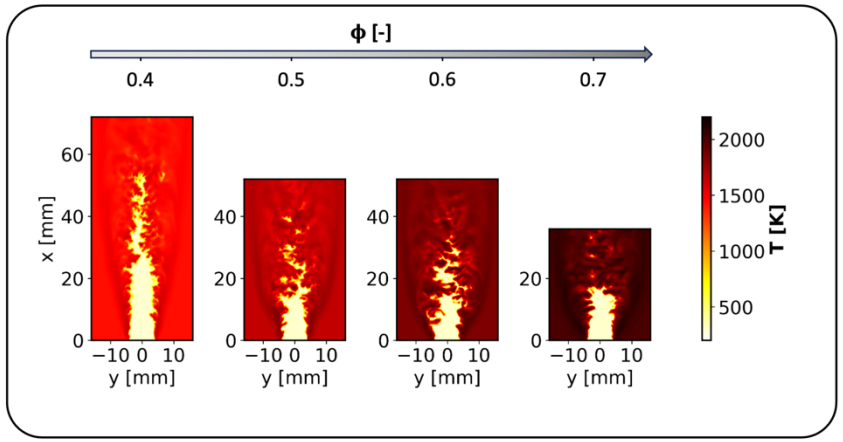
3. Shared **DNS datasets** to train machine learning models are becoming more **diverse** and more **accessible**.



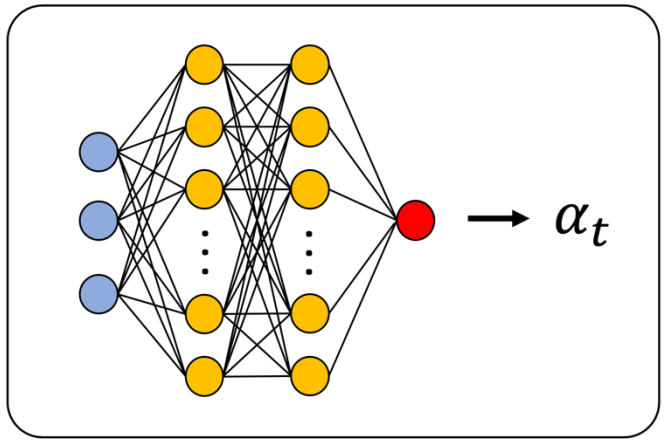
The CYPHER data challenge aims at modeling sub-filter scalar flux of $\tilde{C}$ in lean premixed H2 flames using machine learning

Codabench

1. Training data processing



2. Online model training



3. Closure model evaluation

$$\frac{\partial}{\partial t}\left(\bar{\rho}\tilde{C}\right) + \nabla \cdot \left(\bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}\right) =$$

$$\nabla \cdot \left[\bar{\rho}(\alpha_C + \alpha_t)\nabla\tilde{C}\right] + \bar{\rho}\dot{\omega}_C$$

[8] Malé, Q., Lapeyre, C. J., & Noiray, N. (2025). *Hydrogen reaction rate modeling based on convolutional neural network for large eddy simulation*. Data-Centric Engineering, *6*, e11. doi:10.1017/dce.2025.1

The CYPHER data challenge aims at modeling the sub-filter scalar flux of $\tilde{C}$ in lean premixed H2 flames using machine learning

$$\tilde{C} = \frac{\tilde{Y}_{H_2}^{u} - \tilde{Y}_{H_2}}{\tilde{Y}_{H_2}^{u} - \cancel{\tilde{Y}_{H_2}^{b}}}$$

0

Filtered progress variable

sub-filter scalar flux of $\tilde{C}$

$$\boxed{\boldsymbol{q}_{\tilde{C}} = \bar{\rho}\widetilde{\boldsymbol{u}C} - \bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}}$$

Combustion closure models
(Thickened flame model, Flamelet, FPV, Flame surface density…)

$$\frac{\partial \bar{\rho}\tilde{C}}{\partial t} + \nabla \cdot \left(\bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}\right) + \nabla \cdot \overbrace{\left(\bar{\rho}\widetilde{\boldsymbol{u}C} - \bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}\right)} = \overbrace{\overline{\nabla \cdot (\rho D \nabla C) + \dot{\omega}}}$$

Filtered progress variable **transport equation**

$$\nabla \cdot \left(\bar{\rho}\widetilde{\boldsymbol{u}C} - \bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}\right) \approx \nabla \cdot \bar{\rho}\boxed{\alpha_t}\nabla\tilde{C}$$

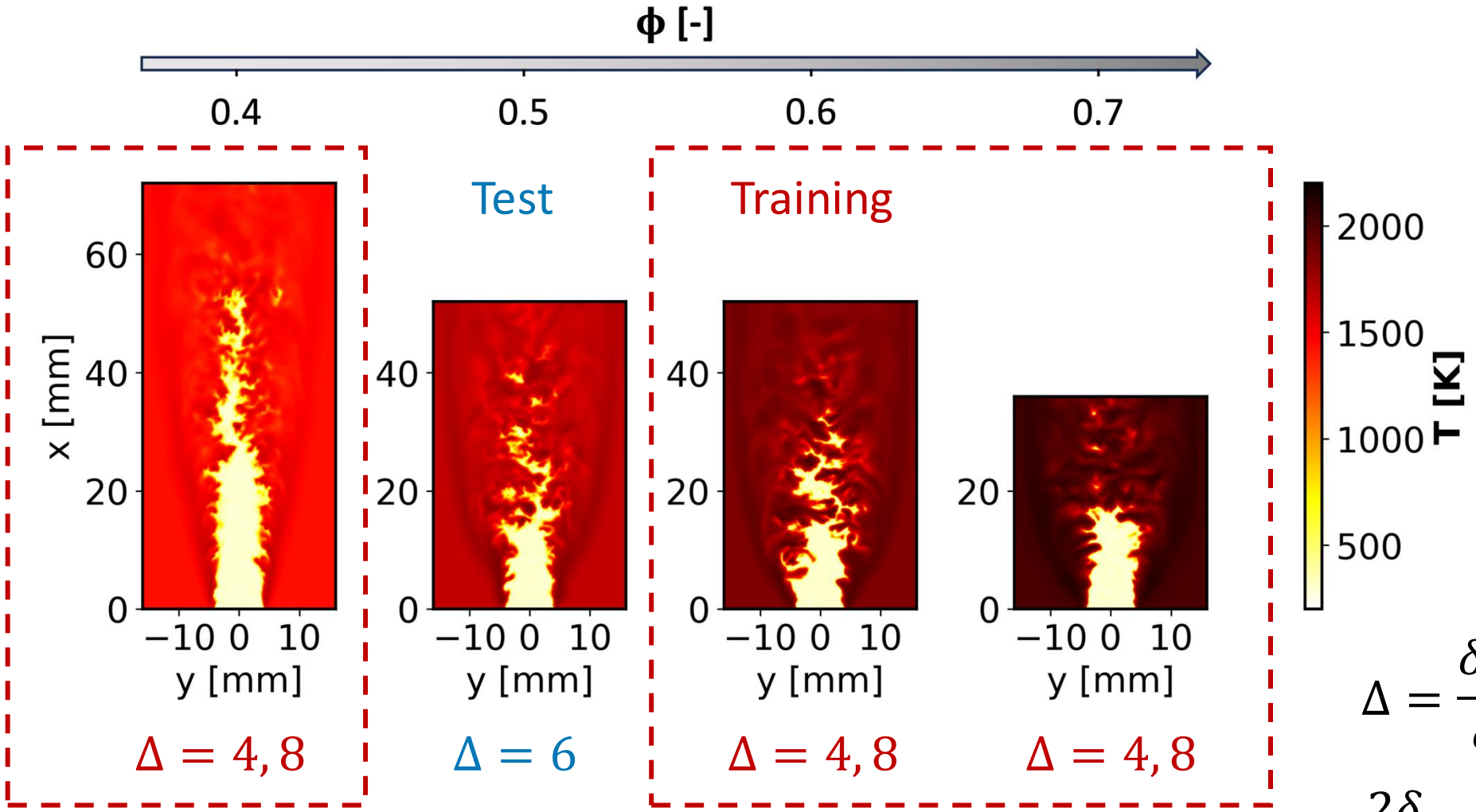
Eddy viscosity assumption
to model the sub-filter scalar fluxes
of the filtered progress variable

$$\boxed{\alpha_t} = \frac{\overbrace{\left(\bar{\rho}\widetilde{\boldsymbol{u}C} - \bar{\rho}\tilde{\boldsymbol{u}}\tilde{C}\right)}}{\bar{\rho}\nabla\tilde{C}}$$

Sub-filter information
available in the **DNS**

Add here a picture of the DNS simulation filtered

The dataset comprises flames at different equivalence ratios and was filtered at various filter sizes



$$\Delta = \frac{\delta_{filter}}{\delta_{DNS}}$$

$$2\delta_{DNS} = \eta_k$$

[8] Malé, Q., Lapeyre, C. J., & Noiray, N. (2025). *Hydrogen reaction rate modeling based on convolutional neural network for large eddy simulation*. Data-Centric Engineering, *6*, e11. doi:10.1017/dce.2025.1

The evaluation metrics includes both the model's accuracy and the inference time

- The **Codabench** platform [9] hosts the competition and allows **automatic evaluation** of the submitted models

Codabench

- The evaluation metrics account for model's **accuracy** and **inference time**

$$\epsilon = \underbrace{\left\| \boldsymbol{q}_{\tilde{C}}^{model} - \boldsymbol{q}_{\tilde{C}}^{DNS} \right\|_2}_{\text{Accuracy}} + \underbrace{\beta_t \, t_{inf}}_{\text{Computational cost}}$$

- β_t is a coefficient to balance the relative weight of the two terms during the evaluation phase

[9] Xu, Z., Escalera, S., Pavão, A., Richard, M., Tu, W. W., Yao, Q., Zhao, H., & Guyon, I. (2022). *Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform*. Patterns (New York, N.Y.), *3*(7), 100543. https://doi.org/10.1016/j.patter.2022.100543

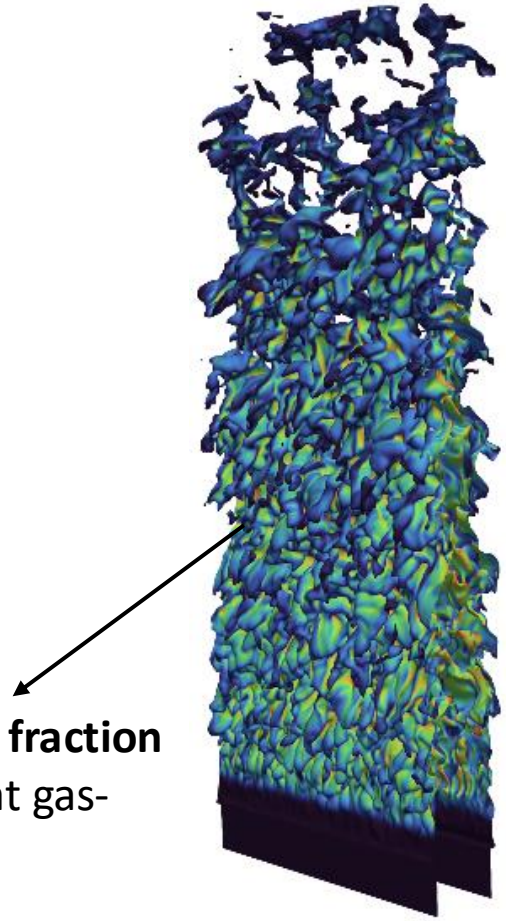
Participants submitted different model architectures, promoting an heterogeneous comparison

Second phase of the challenge: modeling of the unclosed reaction rates in LES of premixed lean hydrogen flames

- Goal:
  Train a **machine learning** model to predict the **filtered reaction rates** $\overline{\dot{\omega}}_{H_2}$

$$\overline{\dot{\omega}}_{H_2} = f\left(\tilde{Y}_{H_2}, \tilde{Y}_{H_2O}\right)$$

- The dataset features a premixed jet flame at conditions that are of interest for practical **gas turbine applications**
- All the interested participants to this community effort are encouraged to **reach out by email** to pasquale.lapenna@uniroma1.it.



Isosurface of **hydrogen mass fraction** coloured with reaction rate at gas-turbine conditions (700 K and 20 atm) [10]

[10] Kassar, S. A., Cantagalli, S., Lauder, W., Arumapperuma, G., & Attili, A. (2026). Turbulent hydrogen premixed flames at high pressure and high temperature [Preprint, presented at the 12th European Combustion Meeting]. arXiv. https://doi.org/10.48550/arXiv.2601.15848

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Bibliography

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https://doi.org/10.1017/dce.2025.1

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[11] Piu, L., Pitsch, H., & Parente, A. (2024). *aPriori: a Python package to process direct numerical simulations*. [Computer software]. GitHub. https://github.com/LorenzoPiu/aPriori